

Does a Music Transformer Know What Key It Is In?

Causal Interventions on an Emergent Tonal World Model

Anonymous ICASSP 2027 submission

Abstract—Music generation models produce convincingly tonal music. Whether they do so by chaining surface statistics or by maintaining an internal representation of musical key that generation *causally uses* has not been tested. We test it directly: a decoder-only Transformer trained on symbolic music from a functional-harmony grammar whose vocabulary contains absolute pitch and time only—no key, chord, or degree tokens—under a pre-registered protocol with frozen decision rules. Our central result is causal: replacing the key component of the residual stream at a single middle layer makes the model continue in the injected key at $5.0\times$ the rate of rank- and norm-matched random edits (guarded strict TKR 0.378 vs. 0.075; significant for 12/12 target keys after Holm correction, replicated across seeds), while staying inside a musicality budget frozen before any edit ran—reaching 58% of the behavioral ceiling obtained by transposing the prompt itself. A capacity sweep sharpens what this means. Next-token accuracy is flat from 0.5M to 86M parameters, yet at 0.5M the key is not decodable beyond the input surface at all, and the same edit moves the key *more* often than at 26M while destroying the music in 90% of cases. Capacity does not buy the key; it buys a key that can be moved without breaking everything else. Finally we leave our own models entirely. On real Bach chorales a *public* 128M checkpoint trained on real music—architecturally identical to one of ours—carries a key state that beats the pitch surface where ours does not, and editing that state redirects the music it composes at $5.7\times$ the rate of matched random edits (11/12 targets, Holm-corrected) with *no* measurable cost to musicality. It also resists: the keys the edit cannot reach are exactly the keys real music rarely uses (Spearman +0.91). Editing a music Transformer’s internal key state changes what key it composes in—in a model people actually use, not only in ours.

Index Terms—music generation, world models, interpretability, causal intervention, tonality

I. INTRODUCTION

Systems that compose music now write convincingly tonal pieces: their outputs settle into a home key, build tension away from it, and return. For listeners this is much of what makes generated music sound like music; for creators it determines whether such systems can be steered.

How they achieve this is largely unexamined. Existing analyses report that musical attributes can be *read out* of a model’s activations by trained classifiers [7]. Reading, however, is not using: a signal can be decodable from

a system’s state without the system consulting it when it acts—a gap documented repeatedly in interpretability research [11]. No study has tested whether a music model *uses* an internal key representation: whether intervening on it, while leaving the input untouched, changes the key of the music the model goes on to generate.

Sequence models can build causal models of the latent state behind their data. A model trained only to predict Othello moves tracks the board those moves imply, and editing that internal board changes which moves it treats as legal [1], [2]. Analogous causal evidence exists for chess, mazes, and RL environments [4], [5], [6], but not for music.

We close that gap. We choose key—tonic and mode—as the state variable for three reasons. First, unlike any candidate in the game domains, key is defined *externally* to the model, by centuries of theory and by psychological measurement [10], so the probed state is not an artifact of our own labeling. Second, among musical state candidates it is the most demanding: metric position is recoverable from time tokens, and harmonic function presupposes a key, whereas key must be *computed* by integrating pitch evidence over time. Third, key admits a natural analogue of Othello’s legal-move rate—diatonic scale membership—so causal success is measurable without a learned judge.

Our contributions:

- 1) A controlled testbed for world-model analysis in music: a leak-free tokenizer, a functional-harmony corpus with exact per-token key labels, and a pre-registered protocol, released as a benchmark (Sec. III).
- 2) Existence and geometry: key is decodable beyond strong input-side baselines in all six trained models, the twelve tonic directions self-organize into a circle of fifths, and—a pre-registered negative result—transposition augmentation does not make the representation a cyclic group (Sec. IV-A).
- 3) Causality: single-layer subspace edits redirect the composed key for 12/12 targets within a frozen musicality budget, and a capacity sweep shows that what capacity supplies is not the key but the ability to move it surgically—dissociating three things the term “world model” usually conflates (Secs. IV-C–IV-F).
- 4) A matched-architecture control on real music: with

TABLE I
TERMINOLOGY. DECISION RULES (DR-*) WERE FROZEN
PRE-EXPERIMENT.

$\kappa(t)$	Key state: tonic ($\mathbb{Z}_{12}$) $\times$ mode at token t
κ^*	Donor key injected by an edit
D-SYN	Synthetic corpus with per-token ground-truth $\kappa(t)$
M-CTRL	Studied model
M-REF	Frozen reference model; used only for the quality guard
Edit	$h \leftarrow h - P_V h + P_V \mu_{\kappa^*}$ in the residual stream
TKR	Rate at which the continuation’s estimated key = κ^*
IKR _{x}	Fraction of continuation pitches diatonic to key x
δ_{PPL}	M-REF perplexity budget, frozen pre-intervention

corpus, protocol, and decision rule held fixed, a public real-trained model clears the input-surface baseline where our synthetic-trained model of the same shape does not, isolating the training data as the variable that decides whether a key state is worth more than the notes themselves (Sec. IV-G).

- 5) The same causal test on that public model: editing its key state redirects the key it composes in ($5.7 \times$ the matched control, 11/12 targets, 100% within the frozen musicality budget), so the world-model claim is not confined to models we built. Its failures are informative—it resists exactly the keys real music rarely uses (Sec. IV-H).

II. RELATION TO PRIOR WORK

Prior work divides along two axes: domain and evidence level. Across domains, world-model analyses have progressed from readability to causal intervention in board games [1], [2], [3], [4], mazes [5], and RL environments [6]. In music, evidence has stayed at the readability level: probing studies decode theory concepts from generation models [7], and steering studies control attributes without testing a state variable [8]. We occupy the empty cell—causal, state-level evidence in music—for a state whose transitions (modulations) themselves follow a grammar, unlike the game domains.

III. METHOD

A. Problem statement and terminology

The input is a symbolic music sequence; the latent state is the running key $\kappa(t)$; the question is whether the generator’s continuation causally depends on its internal estimate of κ . Table I fixes the terms. The pre-registration (supplementary) specifies every decision rule, threshold, and control, all frozen before any experiment ran; deviations are logged and negative results reported.

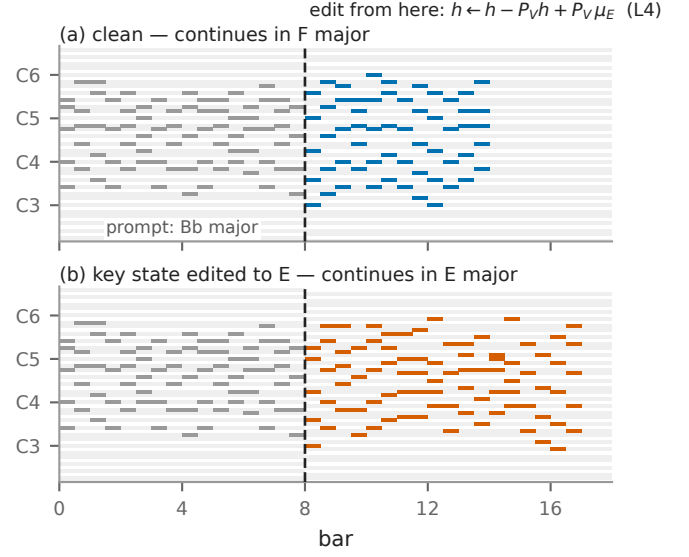


Fig. 1. The claim in one example (real model output, fixed sampling seed [s0]). The same 8-bar Bb-major prompt, continued (a) clean and (b) with the key component of layer-4 activations replaced by the E-major class mean from bar 9 on. **The input does not change**; the continuation’s estimated key follows the injected internal state. Shading marks the scale of the key in force.

B. Pre-registered hypotheses

The protocol, its thresholds and its decision rules were frozen before any experiment ran. We restate the hypotheses in the pre-registration’s own terms rather than paraphrase them, since the decision rules below are only meaningful against the claims they were written for.

H1 (*existence*) Key is decodable from internal activations, and more accurately than from input-surface baselines (recent-window pitch-class histogram / Krumhansl-Schmuckler); the gap widens in ambiguous regions. *Falsified if* no layer’s selectivity-corrected probe beats the input baseline.

H2 (*encoding form*) The key representation is transposition-equivariant: for each transposition T_k there is a linear map R_k in representation space, and $\{R_k\}$ has cyclic-group structure ($R_k \approx R_1^k$).

H2b A model trained with transposition augmentation (R-Aug) shows *stronger* equivariance than one without (R-NoAug)—a manipulation of the training condition. H2b is a sub-claim of H2, which is why no “H2a” appears: the pre-registration numbered the parent claim H2 and its manipulation H2b.

H3 (*causality; the main claim*) A targeted edit of the key subspace moves the key of the continuation to the edit target. The effect significantly exceeds a random subspace edit **matched in rank and norm**, and stays inside the musicality guard’s budget. *Falsified if* TKR is indistinguishable from the control edit, or is achievable only together with quality collapse.

H4 (*persistence and reassertion*) After a *one-shot* edit

the state either (a) persists, (b) is reasserted toward the original key by context, or (c) transitions via a pivot-like passage. The pre-registration assigned H4 to Phase C; its sub-question H4a (does an installed value persist at all, and through what carrier?) became load-bearing for the paper’s framing and was brought forward. Sec. IV-E reports it; the remainder of H4 (pivot structure, annotation) stays in Phase C.

H5 (*localization*) The layers at which an edit is causally effective are confined to a limited band.

H3 asks for a control matched in *rank and norm*, and “norm” admits two readings; we checked both rather than assert the convenient one. The basis reading is satisfied by construction: V and V_{K1} are both orthonormal 512×24 , so $\|V\|_F = \|V_{K1}\|_F = \sqrt{24}$ identically. The stricter reading—that the two edits should *perturb the residual stream equally*—is not imposed by construction, since we never rescale the injected component, so we measured it on real activations during real generation at the tested condition (12 targets, all edited positions). The edit applies $\|\Delta\| = 21.9$ against K1’s 22.1: a ratio of 0.99, or 0.160 vs. 0.162 of $\|h\| = 136.5$, with the *control* marginally the larger. The match is observed rather than enforced, but it is observed, and it is conservative.

That the two agree so closely is itself informative. Both edits project $\mu_{\kappa^*} - h$ onto a rank-24 subspace of $\mathbb{R}^{512}$; a random one captures $\sqrt{24/512} \approx 0.217$ of that vector’s energy, and the key subspace is measured to capture 0.215—no more. The difference between μ_{κ^*} and a given activation is dominated by variation that has nothing to do with key, and the key subspace holds no privileged share of it. So the $5\times$ gap reported in Sec. IV cannot be a magnitude effect: the two interventions are the same size, and only their direction differs.

C. Data and models

To make ground truth exact rather than estimated, all training data are synthetic. **D-SYN**. A functional-harmony state machine ($T \rightarrow S \rightarrow D \rightarrow T$ with substitutions, SATB voicing, optional diatonic melody) generates 200k/10k/10k train/val/test pieces of 278–508 tokens with 0–3 modulations each (pivot-chord, direct, or sequential; fifths-distance stratified), giving every token a ground-truth key label. The three types are planned in equal thirds, but two of them degrade in the token stream: a pivot with no shared diatonic triad emits a direct jump, and a sequential shift over an odd fifths-distance cannot be halved into two equal steps. The realized corpus is therefore 18.6% pivot, 16.5% two-step sequential, and 64.9% single jumps—a fact the surface baselines benefit from, not our probe. We also note that our “sequential” is a documented simplification (two equal-interval shifts one bar apart), not a full melodic sequence. Every token still carries an exact key label. **Tokenizer**. To prevent state leakage the vocabulary’s 124 types are **BAR**, **POS**, absolute-MIDI **PITCH** and **DUR** (121 types) together with

the three structural symbols **PAD/BOS/EOS**, which are key-invariant. Nothing names a key, chord or degree, and a unit test enforces this. **Models**. M-CTRL is a GPT-2 style decoder (8 layers, 8 heads, $d=512$, context 512, $\sim 25M$ parameters), trained 12k steps under two regimes—with (R-Aug) and without (R-NoAug) uniform 12-fold transposition augmentation— $\times 3$ seeds. M-REF shares the architecture but uses a disjoint seed *and* data split, and serves only as the musicality guard, avoiding circular evaluation. All models pass a pre-frozen quality gate (validation next-token top-1 0.878–0.881 vs. a 0.434 constant-predictor rate; unconditional in-key ratio 0.964–0.977 vs. a 0.608 threshold).

D. Phase A: probing and geometry

To separate internal computation from input transcription, every probe result is reported against three control families. Linear (multinomial logistic) probes map residual-stream activations $h_\ell(t)$ to $\kappa(t)$ (24 classes) with sequence-level splits. **C1** permutes key identities per sequence—a control task in the sense of [11]; **C2** is an untrained model; **C3** are input-side baselines at the same positions (pitch-class histogram logistic regression, and Krumhansl–Schmuckler (KS) template matching; windows $W \in \{8, 16, 32, 64\}$). DR-H1: the probe supports H1 iff its selectivity-corrected F_1 exceeds the best C3 baseline with a sequence-level BCa 95% CI excluding zero in some layer. Ambiguity is stratified by KS posterior entropy ($W=16$). For geometry, orthogonal Procrustes maps R_k are fit between activations of pieces and their k -semitone transpositions, and a cyclicity error tests group structure:

$$\varepsilon_{\text{cyc}} = \frac{1}{10} \sum_{k=2}^{11} \frac{\|R_k - R_1^k\|_F}{\|R_k\|_F}, \quad (1)$$

normalising each term by $\|R_k\|_F$ before averaging, as pre-registered. The sum starts at $k=2$ because $R_1 - R_1^1$ is identically zero and would only dilute the mean, and stops at $k=11$ because T_{12} is the identity; R_1 defines the powers. DR-H2b: $\varepsilon_{\text{cyc}}(\text{R-Aug}) < \varepsilon_{\text{cyc}}(\text{R-NoAug})$ (one-sided Wilcoxon across seeds).

E. Phase B: causal intervention

To test *use* rather than presence, we edit the state and score the behavior. From bar 9 of a key-stable 8-bar prompt, a sustained edit replaces the key component at one layer, $h \leftarrow h - P_V h + P_V \mu_{\kappa^*}$, with V an orthonormal key subspace and μ_{κ^*} the class-conditional mean of the donor key. “Sustained” is meant literally and asymmetrically, and the asymmetry is worth stating because it is easy to assume otherwise: the edit is applied at *every* token position from t^* (the bar-9 boundary) onward—**BAR**, **POS**, **PITCH** and **DUR** alike—and re-applied at every generation step, whereas the *probe* is trained and scored at sampled positions, uniformly over token types (the predict-pitch reading convention enters only with the public-model study, Sec. IV-G). (What a *one-shot*

installation leaves behind—and through what carrier—is Sec. IV-E.) We write to the whole stream and read from one position in it. Restricting the edit to pitch-choosing positions is a natural variant and we have not run it. Subspaces: **V-PROBE** (probe row space, rank 24), **V-MEAN** (class-mean span), **V-DAS** (interchange-trained, ranks 8/24). Controls: **K1** rank- and norm-matched random subspace (Sec. III-B); **K2** sham edit, required bit-identical to the clean run (passed, 100/100 prompts); **K3** a different layer’s subspace; **K4** behavioral reference, transposing the prompt itself. Continuations (16 bars; 12 major targets $\times$ 100 prompts $\times$ 8 layers; sampling noise paired with a clean twin) are scored by KS key estimation, IKR, and a grammar guard. TKR is reported *strict* (the estimated key equals κ^* exactly) and *tolerant*. The pre-registration defines tolerant as the disjunction of three relations, not fifths adjacency alone: fifths distance ≤ 1 *with the same mode*, the relative major/minor (mode differs, tonics 3 or 9 semitones apart), or the parallel key (mode differs, same tonic). Every headline number in this paper is strict; tolerant exists to bracket the estimator’s known confusions (Sec. V), not to soften the claim. The budget $\delta_{\text{PPL}} = 0.613$ (90th percentile of the M-REF perplexity rise across 7,989 natural modulations in validation data) was fixed *before* any edit ran; edits exceeding it do not count as successes. DR-H3 requires $\text{TKR}(\text{edit}) > \text{TKR}(\text{K1})$ with Holm-corrected Wilcoxon $p < .05$ in $\geq 8/12$ targets at the best in-budget condition.

IV. RESULTS

Intervention numbers are from the R-Aug seed-0 sweep [s0] unless noted; probing holds for all six M-CTRL models, and the seed-1 sweep replicates every Phase-B verdict (Sec. IV-F).

A. H1: the key is decodable beyond the input surface

The key is *computed* by the model, not transcribed from the input: the corrected probe–baseline margin excludes zero in layers 1–7 of every trained model (peak +0.105). It fails only at L0, and fails there in the informative direction: the margin is significantly *negative* (−0.20, CI [−0.211, −0.182]), i.e. at the raw embedding the input surface outperforms the model’s own representation of it. Table II gives the numbers at the layer the edit acts on.

1) The history-length confound, and what survived it:

The comparison above has an asymmetry that could have explained it entirely: the probe reads $h_\ell(t)$, which attends over the full prefix (≤ 512 tokens, ~ 170 notes), while C3 sees a W -token window with $W \leq 64$ (~ 21 notes)—and the best C3 sat exactly at $W=64$, the *edge* of the pre-registered sweep, with its F_1 still rising steeply (0.686 at $W=32$). Both the headline margin and the ambiguity contrast could then be read as “the probe sees more history,” not “the model computes a state.”

We therefore reran C3 at the same positions with the sweep extended to the full prefix

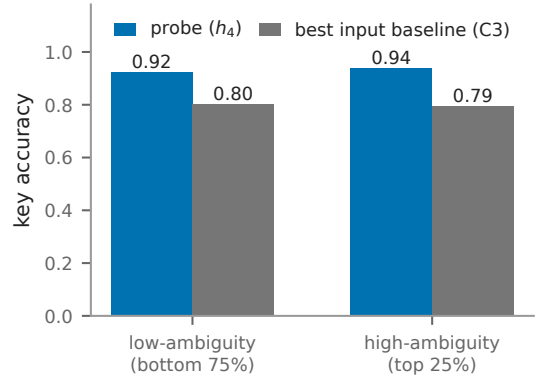


Fig. 2. A cautionary result. Against the *pre-registered* baseline (shown), the probe’s advantage concentrates in the top ambiguity quartile (0.94 vs. 0.79)—which reads as “the state earns its keep where the surface fails.” Against the strengthened baseline of Sec. IV-A1 the advantage is uniform across strata (+0.097 in both): the concentration was an artifact of the $W=64$ window, itself weakest where a short window is uncertain.

($W \in \{96, 128, 192, 256, 512\}$; 512 exceeds the longest piece) and with two post-hoc variants that only strengthen the baseline: exponentially recency-decayed full-history histograms (half-life $\lambda \in \{32, \dots, 256\}$ tokens) and a concatenation of short and full windows $[W16|W512]$. History alone does not close the gap. The plain-window curve turns over *inside* the extension, at $W=96$ (0.801), and falls monotonically to 0.752 at the full prefix: with 0–3 modulations per piece, evidence from before a modulation poisons a histogram that cannot down-weight it, which is precisely the estimation problem an internal state solves. The strongest surface overall is the concatenation, at 0.824; the decayed variant peaks at 0.814 ($\lambda=32$). Against that strongest baseline the corrected margin at L4 is +0.071 (BCa 95% CI [0.056, 0.081]), and every DR-H1 verdict in this paper—6/6 main-line support, both 0.5M failures, both 3.4M and both 86M passes—is unchanged, though the 3.4M margins thin to +0.015 [0.004, 0.023] and +0.012 [0.001, 0.021].

One registered sub-claim does *not* survive. Against the pre-registered baseline the probe’s advantage concentrated where KS estimation is uncertain (0.936 vs. 0.795 in the top ambiguity quartile); against the strengthened baseline the advantage is *uniform*—+0.097 in the top quartile, +0.097 elsewhere. The apparent ambiguity-specific edge was an artifact of the $W=64$ cap, which is weakest exactly where a short window is uncertain. We report H1’s “the gap widens where the key is ambiguous” as not supported in its strengthened form, and Fig. 2 should be read with that caveat.

B. H2: fifths geometry; equivariant but not a group

The representation carries music-theoretic structure, but not the strictest form we pre-registered. Projected to two principal components, the twelve major-tonic directions order as C–G–D–A–E–B–F♯–... (Fig. 3): circle-

TABLE II

KEY DECODING AT L4 (R-AUG SEED 0), 24-CLASS MACRO- F_1 ON HELD-OUT SEQUENCES; DR-H1 SUPPORTED IN 6/6 MODELS. NOTE THE TWO C1 ROWS MEASURE DIFFERENT THINGS: A CONTROL PROBE TRAINED ON PERMUTED LABELS BARELY LEARNS THEM (0.040, NEAR THE 1/24 CHANCE RATE—THE CONTROL TASK IS GENUINELY UNLEARNABLE), AND PREDICTS THE TRUE KEY AT 0.032—THE SELECTIVITY FLOOR THE MARGIN SUBTRACTS. THE MARGIN RECONCILES EXACTLY: $0.927 - 0.032 - 0.790 = 0.105$.

Method / baseline	macro- F_1
Linear probe on $h_4(t)$	0.927
MLP probe (secondary)	0.932
Best input baseline (C3: PC-hist LR, $W=64$)	0.790
Untrained model (C2)	0.243
C1 control probe, on its own permuted labels	0.040
C1 control probe, on the true key (selectivity floor)	0.032
Corrected margin, BCa 95% CI [0.091, 0.114]	+0.105

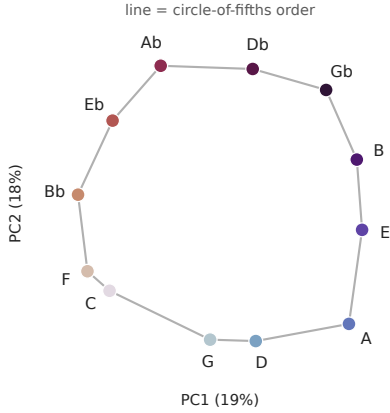


Fig. 3. The twelve major-tonic probe directions (L4, first two PCs) self-organize into a circle of fifths. No key, chord, or degree token exists in the training vocabulary.

of-fifths neighbour agreement 1.00 vs. 0.00 for the chromatic ordering, at every probed layer (exploratory). Transposition acts as a generalizing linear map: Procrustes R_k fit on half the sequences aligns held-out activations with 0.05–0.11 relative error. The family is not cyclic, however: $\varepsilon_{\text{cyc}} \approx 0.98$ vs. ≈ 1.41 for random orthogonal maps [Eq. (1)]. Our pre-registered comparison is negative: augmentation does not improve cyclicity (3 seeds per regime; R-Aug 0.983 vs. R-NoAug 0.981; one-sided $p = 0.83$), so DR-H2b is *not* supported. The representation is transposition-equivariant in effect but not organized as a cyclic group, and the key diversity already present in the data appears to suffice.

C. H3: editing the key state changes the composed key

Editing the internal key redirects generation. At the best in-budget condition (V-PROBE, L4), $\text{TKR}(\text{edit}) - \text{TKR}(\text{K1}) > 0$ holds for 12/12 targets (Holm-corrected $p < 10^{-4}$ each; rank-biserial $r = 0.64$ –1.00) against a pre-registered bar of $\geq 8/12$: DR-H3 is supported. Fig. 4

gives all conditions. Pitch content triangulates the same conclusion without the estimator: $\text{IKR}_{\kappa^*} = 0.937$ vs. $\text{IKR}_{\kappa_{\text{src}}} = 0.646$ after editing. Success falls gently with target distance on the circle of fifths (0.395 at distance 1 to 0.280 at the tritone), while random edits collapse to zero once the identity target is excluded. The edit reaches 58% of the behavioral ceiling (K4), quantifying an *actionability gap*: the state is readable in a rank-24 subspace of one layer, yet a single-layer edit realizes only part of the behavioral effect. The fraction depends on a convention we should state. K4’s perplexity excess is measured against the *untransposed* clean twin, so it scores the transposition rather than any damage, and guarding K4 is not well defined. We therefore divide the edit’s guarded rate (0.378) by K4’s raw rate (0.652), giving 58%. Applying the guard to both sides gives $0.378/0.614 = 62\%$ and to neither $0.429/0.652 = 66\%$; we report the smallest of the three, which is the one that flatters us least. Two subsidiary findings: the *optimized* subspace (V-DAS) underperforms the probe’s readout directions (Sec. V), and V-MEAN edits destroy output quality at deep layers (guard pass $\leq 5\%$ at L4–L5)—the guard is load-bearing, a point Sec. IV-F makes decisive.

The blanket write itself—every token position, one type-averaged μ_{κ^*} —is a design one should suspect of damaging the music needlessly, so we tested it: the same condition with the write restricted by token type. It is vindicated, and the refutation is informative. Restricting to PITCH positions is causally *null* (guarded 0.077, at the K1 floor) while leaving the music essentially untouched (guard pass 0.974); restricting to POS+PITCH recovers 0.156; the *complement*—BAR and DUR positions only—recovers 0.261, more than POS+PITCH from fewer positions; and the two halves sum approximately to the whole (0.156+0.261 vs. 0.429 raw). Guard rates do not improve under any mask, and the perplexity cost per unit of guarded TKR is *lowest* for the blanket write: the damage tracks the effect, not the write’s breadth. The causally potent component of the key state thus sits mostly at bar-line and duration positions, not at the pitch emissions themselves—a third dissociation alongside Sec. IV-D’s, whose mechanism we leave open: probe F_1 by token type is flat, per-type edit displacement is comparable, and attention at the reading layers concentrates on PITCH sources, so none of the three obvious accounts (readability, edit magnitude, attention anchoring) explains it.

D. H5: readability is diffuse; causal efficacy is sharp

The layer profile of the edit-minus-control effect is single-peaked (0.03, 0.04, 0.19, 0.26, **0.30**, 0.26, 0.23, 0.23 for L0–L7); the best-vs. -median bootstrap CI [0.038, 0.109] excludes zero, so DR-H5 is supported. The two profiles in Fig. 5 are *not* the same shape, and the difference is the finding. The probe saturates across L2–L6 (macro- F_1 0.920–0.929, all within 0.01 of its maximum), so the key is legible from most of the network;

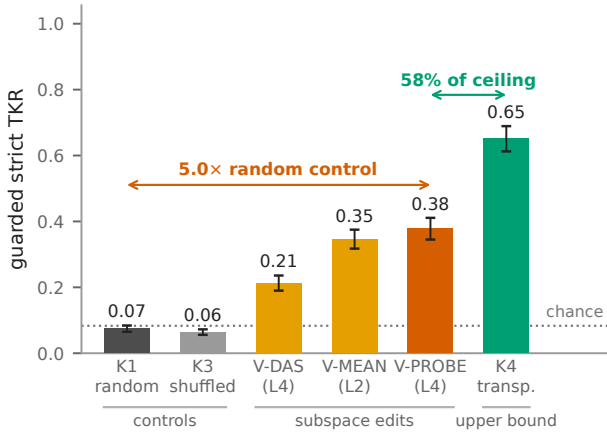


Fig. 4. Guarded strict TKR across conditions [s0]: the continuation lands exactly on the injected key *and* stays within the frozen musicality budget. Chance = 1/12. Bars: prompt-level bootstrap 95% CI.

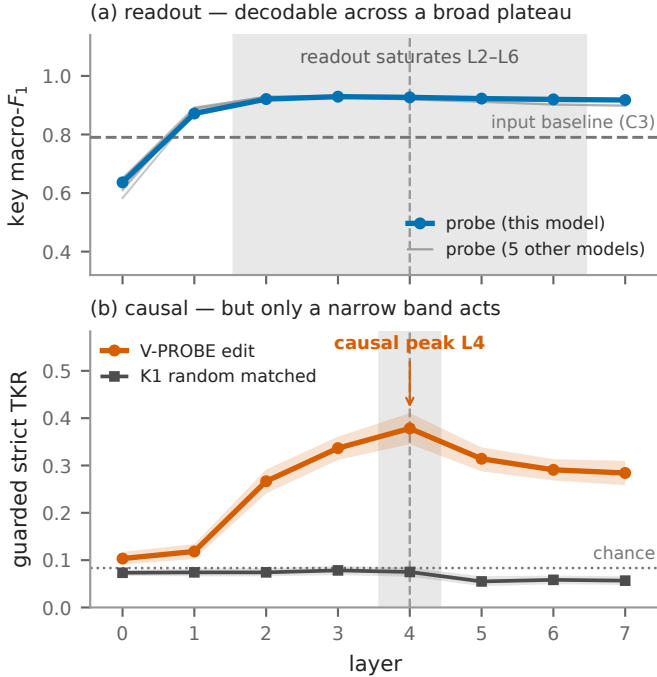


Fig. 5. (a) Probe macro- F_1 per layer, all six M-CTRL models (thin lines; R-Aug seed 0 highlighted): the key is legible across a broad plateau. (b) Causal effect [s0]: guarded strict TKR of the V-PROBE edit vs. the K1 matched random control (bands: prompt-level bootstrap 95% CI). Only a narrow band around L4 moves the output.

yet a single-layer edit moves the output only near L4, and barely at all at L0–L1. Where a state is legible is a weak guide to where it is used—a caution for interpretability work that localizes a computation by probing accuracy alone.

E. H4a: the variable is re-estimated, not stored

Everything above edits under a *sustained* clamp, and a clamp cannot witness the “install a value” semantics our framing borrows: perhaps the model merely tolerates being steered, and maintains nothing. So we installed the value *once* and let go. With no key–value cache the past is recomputed at every step, so “once” means: the edit is pinned to the positions of the first generated bar (each row’s window closes at its own next bar line) and no later position is ever touched; later positions may attend to the pinned bar-9 state and to the notes it caused. Sustained, clean, and a K1 one-shot bracket it; all conditions share sampling randomness, and the one-shot’s first bar is *identical* to the sustained run’s by construction ($\text{IKR}_{\kappa^*} = 0.805$ in both, an internal validity check).

Released, the effect does not decay—and it does not complete (Fig. 6a). IKR_{κ^*} drops once ($0.805 \rightarrow 0.685$) and then holds a flat plateau (0.65–0.70) for thirteen bars, +0.12 above the K1 floor of 0.55 (average scale overlap between random major keys). The music neither returns to the source key nor settles in the target: per-bar KS names the target in only ~9% of bars (the clamp climbs to 28%), and under a strict last-four-bars metric the continuation returns to the source in 31% of runs—*less* often than after a matched random perturbation (45%), which itself matches the clean run’s own 47%: a random one-bar perturbation derails nothing; the installed bar durably shifts the tonal center of mass without finishing the modulation.

Then the control that decides what “persists” means (Fig. 6b): splice the one-shot run’s bar-9 *tokens* after the clean prompt and continue with no activation edit anywhere. The splice reproduces the plateau in full (0.687 vs. 0.671 across bars 2–14; state pinning adds -0.016 , i.e. nothing). **The carrier is the music, not the pinned state:** a value installed in the activations persists exactly insofar as it writes itself into the notes, which then stand as ordinary input evidence. The key variable this paper’s probes read and this paper’s edits move is a continuously re-estimated statistic of the evidence—causally consulted at every step (H3), never latched (H4a). Our framing must therefore not rest the “set a variable” distinction on persistence-after-release, and does not: it rests on the remove-and-install semantics, the matched controls, and the dissociations of Secs. IV-C–IV-D.

F. Replication, and what capacity actually buys

Second seed. An independently trained R-Aug seed-1 model reproduces every Phase-B verdict: DR-H3 supported with V-PROBE significant in 12/12 targets (TKR 0.23–0.41 vs. K1 0.02–0.10), and DR-H5 supported (CI [0.122, 0.181]). Its profile peaks at L2 rather than L4: the *localization* replicates, its exact depth does not, and we report both profiles rather than a pooled one.

Capacity. Training the same data and protocol at 0.5M (L2, $d=128$), 3.4M (L4, $d=256$), 26M (main line), and 86M

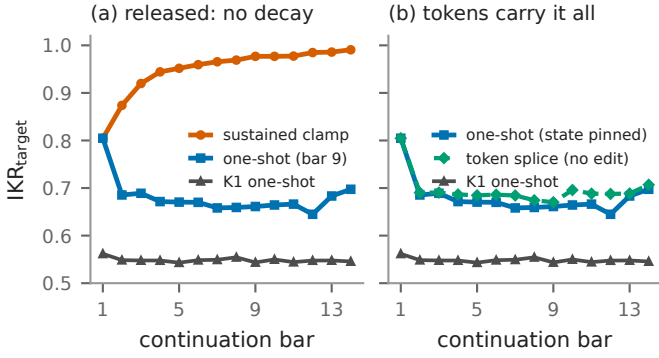


Fig. 6. H4a. (a) Released after a one-bar installation, the pitch-content shift neither decays nor completes: a flat plateau +0.12 over the K1 floor for 13 bars. (b) Splicing the installed bar’s *tokens* with no activation edit reproduces the plateau; pinning the state adds nothing. The variable is re-estimated from evidence, not stored.

(L12, $d=768$) parameters (counted from the checkpoints) dissociates three things the term “world model” usually conflates (Fig. 7). *Behavior* is flat: next-token top-1 is 0.868–0.879 across a $170\times$ capacity range. *Readability* is not: at 0.5M the probe peaks at F_1 0.63–0.66, *below* the input baseline (0.79), and DR-H1 is not supported in either seed—a behaviorally equivalent model that has no key state at all, which is the surface-statistics account made flesh, and which our pre-registered test correctly refuses to credit. (All size verdicts are unchanged against the strengthened baseline of Sec. IV-A1; the 3.4M margins thin to +0.015 [0.004, 0.023] and +0.012 [0.001, 0.021]—the state first appears just barely above the strongest surface.) *Manipulability* is a third thing again. At 0.5M the edit changes the key *more* often than at 26M in raw terms (0.55 vs. 0.43) and passes the musicality guard only 10% of the time (vs. 80% at 26M): with two layers, “editing the key subspace” is a blunt overwrite of the output, not the movement of a separable state. Reporting guarded TKR alone would hide this; reporting raw TKR alone would credit a surface model with a world model. **Capacity does not buy the key—it buys a key that can be moved without breaking everything else**, and the pre-registered guard is precisely the instrument that tells the two apart.

Manipulability does not simply keep improving, however. It peaks near 26M and then *falls*: the 86M model reaches guarded TKR 0.18, about half the 26M value. We suspected an artifact of our own making—each size model was edited at its probe- F_1 peak, a heuristic this paper itself refutes—so we swept all twelve layers of the 86M model. The hypothesis was wrong: its causal peak (L6, 0.206) is statistically indistinguishable from the layer we had already used (L3, 0.202). The decline is real. A natural reading, which we did not test and do not claim, is that greater depth leaves more downstream layers able to re-derive the key from the surviving context, so a single-layer edit is more easily repaired—the actionability gap

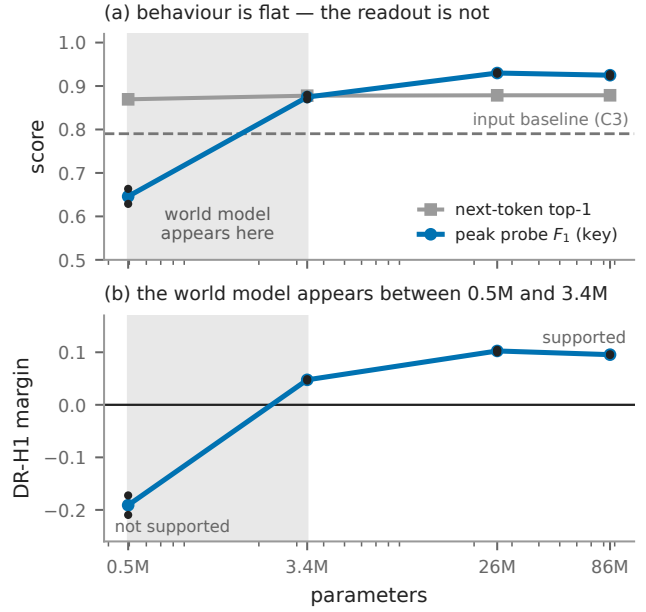


Fig. 7. Behavioral equivalence, representational dissociation. Across a $170\times$ capacity range next-token accuracy is flat, but the key readout is not: below ~ 6 M the probe cannot beat the input baseline and the pre-registered DR-H1 margin is negative. Dots: seeds.

widening with scale rather than closing. (Layer sweep: one seed.)

G. Real music, and a public model: what generalizes

Two questions remain for a synthetic-data study: does the key readout survive real music, and does any of this describe a model people actually use? We answer both with one matched comparison, on 300 Bach chorales whose *local* key at every note is given by published Roman-numeral analyses.¹

Our model does not clear the surface on real music. Probed on the chorales, our 86M model reaches macro- F_1 0.534 against a selectivity floor of 0.035—the key subspace is real and transfers, it is not an artifact of our grammar—but the best input baseline (KS, $W=64$) reaches 0.451, and the corrected margin, +0.034 (CI [−0.021, 0.103]), does not exclude zero. DR-H1 is *not* supported out of distribution.

A public model trained on real music does clear it. The Anticipatory Music Transformer [9] (Apache-2.0; Lakh MIDI, MetaMIDI, FMA, and 450k commercial records) is, at its *small* size, *architecturally identical* to our 86M model: 12 layers, $d=768$, and 85.1M non-embedding parameters against our 85.2M. Only the totals differ (128M vs. 86M), and only because its vocabulary holds 55k tokens where ours holds 124—so the stacks being compared are the same size, and what differs is what they were trained on. Its vocabulary is (arrival-time, duration,

¹Scores: **bach-370-chorales** (CC BY-NC-SA); analyses: *When in Rome* (CC BY-SA). Neither is redistributed.

note) triples—leak-free in exactly the sense ours is. We probed it on the same chorales, at the same positions, with the same controls and the same decision rule; the two C3 baselines land at 0.451 and 0.454, confirming the setups are matched. It reaches macro- F_1 0.690 (L10) with a corrected margin of +0.196 (CI [0.132, 0.260]): **DR-H1 supported**. With architecture, corpus, protocol, and decision rule held fixed, *the training data is the single free variable*, and it is what decides whether a key state beats the pitch surface. The layer profiles say the same thing from the other side: our synthetic model’s readout peaks early (L2) and decays to 0.17 by L11 on this out-of-distribution input, while the real-trained model’s dips mid-network and *builds* to its peak at L10.

One caution, and it is a large one. The probe must be read where the model is about to *choose* a pitch (the token before a note). Read at the note itself—after the pitch is committed, when the model is predicting the next onset—the same public model, on the same chorales, with the same probe, controls and rule, produces a different picture and a different verdict. Its layer profile inverts from a U-shape peaking deep (F_1 0.618 at L0, 0.357 at L4, 0.690 at L10) to a monotone decay (0.545, 0.227, 0.093), and the corrected margin collapses from +0.196 (CI [0.132, 0.260]) to +0.042 (CI [−0.034, 0.092]), which does not exclude zero: read at the note, **DR-H1 is not supported** on this model. Nothing changed but the position. Had we probed where the pitch has already been emitted, we would have concluded that a public music Transformer’s key state does not beat the pitch surface and that it fades with depth—and we would have published it. Where one probes is not a free parameter, and we report the negative convention alongside the positive one rather than only the one that worked.

H. The public model uses its key state

Readability is not use, and the whole point of this paper is that the difference is testable. We therefore put the public model through the same intervention its synthetic counterpart faced. The subspace is the row space of the probe we fit on it; the guard is a *different* public checkpoint (music-medium refereeing music-small: different weights, so not circular), with δ_{PPL} frozen at 0.849 nats—the 90th percentile of that reference’s perplexity rise across 1,180 *natural* modulations in the chorales. Because this paper shows the probe peak is not the causal peak, the edit layer cannot be inherited from the probe: it is searched on 20 chorale prefixes and then judged on 60 prefixes it never saw. The sham edit is bit-identical to the clean run (20/20), so any movement is the edit’s.

We state the freeze order precisely, because it differs from the main experiment’s. For our own models the budget was frozen before *any* edit ran. Here it was frozen after the 20-prefix layer scan and before the held-out evaluation—the order the ledger records. The scan never consults the budget (it ranks layers on unguarded

TKR(edit) – TKR(K1)), and every number we report comes from the evaluation that followed the freeze; but the scan’s raw outcomes were visible when the budget’s parameters were fixed, so this is a weaker guarantee than the pre-registered one and we do not claim otherwise.

DR-H3 is supported (Table III). At L8, on held-out prompts, the edit lands the continuation on the injected key at $5.7\times$ the rate of the matched random control (0.365 vs. 0.064), significant in 11/12 targets after Holm correction. IKR crosses: the continuation is *more* diatonic to the injected key than to the prompt’s own (0.824 vs. 0.785). And the edit costs the music nothing measurable—100% of edited continuations sit inside the frozen budget (median excess +0.125 nats against a budget of 0.849), where our own model managed 87%. A key state learned from real music is *more* cleanly separable, not less. One consequence deserves stating plainly, since the guard does discriminating work elsewhere in this paper: here it does none. Both the edit and K1 pass it at 100%, so guarded and raw TKR coincide exactly (0.365 and 0.064) and the word “guarded” in Table III filters nothing. That is the finding, not a gap in it—but the reader should not picture the budget separating survivors from casualties on this model, because there are no casualties.

Two findings here are only visible on a real model. First, the causal peak (L8) is again *not* the probe peak (L10): the diffuse-readout/sharp-effect dissociation of Sec. IV-D is not an artifact of synthetic data. Second, **the model has a key prior and resists being pushed out of it**. The single target that fails outright is F $\sharp$ major (TKR 0.000), and across all twelve targets success is almost perfectly ordered by how common that tonic is in the chorale corpus. We count key frequency two ways over the same 300 chorales, both from the human Roman-numeral analyses. Weighting each key by how much music is actually spent in it (note-level local labels, $n=138,249$) gives Spearman +0.91 against TKR; counting each chorale’s opening key once ($n=300$) gives +0.75. With only twelve tonics the asymptotic test is unreliable, so we report one-sided Monte-Carlo permutation p (direction predicted a priori): $p < 10^{-4}$ and $p = 0.003$ respectively. On the note-weighted count, D at 13.8% of the corpus reaches TKR 0.600 and G at 21.4% reaches 0.550, while D $\flat$ at 0.2% reaches 0.233 and G $\flat$ at 2.2% reaches zero.

Two limits of this analysis deserve naming. The chorale corpus is a *proxy* for which keys Western tonal music favours; it is not this model’s training distribution, which is Lakh MIDI, MetaMIDI and FMA [9]. The correlation therefore establishes that the model has a key prior aligned with tonal practice, not that the prior was inherited from any particular corpus—identifying its source would require the training distribution itself. And G $\flat$ is the one target where the prior argument and the estimator’s known weakness coincide: it is both the rarest sharp-side key and the enharmonic seam of the circle, so its zero should be read as the ordering’s endpoint, not as independent

TABLE III

EDITING THE KEY STATE OF A *PUBLIC* MODEL TRAINED ON REAL MUSIC (MUSIC-SMALL, 128M, APACHE-2.0), ON 60 HELD-OUT BACH CHORALE PREFIXES. THE EDIT LAYER WAS CHOSEN ON A DISJOINT SET OF PROMPTS. GUARD: A DIFFERENT PUBLIC CHECKPOINT, BUDGET FROZEN BEFORE ANY EDIT.

Guarded strict TKR, V-PROBE edit (L8)	0.365
Guarded strict TKR, K1 matched random control	0.064
ratio	5.7×
Targets significant after Holm correction	11/12
Guard pass rate (edit)	100%
IKR to the injected key vs. to the prompt’s key	0.824 vs. 0.785

TABLE IV

K3 (LAYER-SHUFFLED SUBSPACE) RESOLVED BY LAYER. K3 INJECTS LAYER $\ell+4$ ’s SUBSPACE AT LAYER ℓ ; GUARDED STRICT TKR, R-AUG SEED 0. THE LAYER-POOLED MEAN (0.160) AVERAGES TWO OPPOSITE REGIMES AND DESCRIBES NO CONDITION THAT WAS RUN.

inject at (borrowed from)	L0 (L4)	L1 (L5)	L2 (L6)	L3 (L7)	L4 (L0)
K3	0.003	0.274	0.387	0.313	0.064
own edit	0.103	0.118	0.267	0.337	0.378
K1	0.073	0.074	0.074	0.078	0.075

evidence. Our synthetic model, whose key distribution is uniform by construction, could not have shown any of this.

V. DISCUSSION AND LIMITATIONS

The three levels support the world-model reading *within this training condition*: a Transformer trained on notes alone maintains a mid-layer key variable with music-theoretic geometry, and generation consults it. The gap to the behavioral ceiling and the failure of edits at L0–L1 indicate a distributed, redundantly re-estimated state rather than a single canonical register—a contrast with Othello-GPT’s near-complete editability. Sec. IV-E turns the natural attribution into a measurement: music continually re-grounds tonality in the immediately preceding notes, and accordingly an installed key value survives only through the notes it writes—the state is re-estimated, never latched.

K3 deserves its own paragraph, because its layer-pooled value misleads and we had initially read it wrongly. Pooled over layers K3 is 0.160, which sits invitingly “between K1 and the true edit” and reads as a graded partial effect. Resolved by layer (Table IV) it is nothing of the kind. At the tested layer L4, K3 borrows L0’s subspace—the one layer whose probe fails to beat the input surface—and accordingly does nothing (0.064, at or below K1’s 0.075). That is the clean negative control the design intended, and it says the L4 effect requires L4’s own subspace. Everywhere else the result inverts: subspaces borrowed from L5–L7 and injected at L1–L3 reach 0.274–0.387, and at L2 the borrowed subspace (0.387) outperforms that

layer’s own edit (0.267) and even the L4 headline. The key subspace is thus largely *shared* across the middle stack rather than re-learned at each depth; what depth governs is not which direction encodes the key but whether writing to it changes the output. K3 is a valid negative control at L4 only, and we no longer read its pooled mean as evidence of anything.

Five observations bound our claims. *Mode*: the pre-registration (SPEC §4) fixed the injected mode to major and designated minor targets a secondary condition. They have not been run. Every causal claim in this paper is therefore a claim about major targets, and whether a minor key state is equally manipulable—or whether the major/relative-minor distinction is even carried in the same subspace—is open. *V-DAS below V-PROBE*: the optimized subspace may simply be undertrained relative to its search space; we report it as observed and release its configuration, but do not claim optimization is inherently inferior. *Estimator*: the KS judge confuses fifth-neighbours, which strict/tolerant reporting brackets and IKR triangulates. *Capacity*: the sweep varies depth and width together, so it locates a threshold, not its cause. *External validity*: Sec. IV-H carries both the readout and the causal result to a public model trained on real music, so the world-model claim is no longer confined to our own synthetic setting. What remains ours alone is the *exactness* of the ground truth: only the synthetic corpus gives a per-token key label that no estimator had to guess. Whether edited continuations modulate *musically* (pivot-like transitions vs. jumps) is measured in the journal version. All training data are synthetic; no artist recordings or copyrighted corpora are used for training, and the evaluation corpora are used under their stated licenses.

VI. CONCLUSION

We gave pre-registered causal evidence that a music Transformer tracks musical key as an internal state variable that generation consults, and quantified how far a single-layer edit goes toward full behavioral control. The capacity sweep separates three properties usually bundled under “world model”—the state being present, being legible, and being separately manipulable—and shows they emerge at different points. The testbed (exact state labels, leak-free vocabulary, frozen decision rules) is released as a benchmark. Because music offers state variables that theory defines and ears can verify, we expect it to be a useful substrate for asking what sequence models know; our next question is whether models also internalize how keys are *supposed* to change.

Reproducibility. Anonymized code, data generator, pre-registration, and an audio demo page (clean vs. edited continuations): <https://anonymous.4open.science/r/TBD>. [TBD (run required): create before submission.]

REFERENCES

- [1] K. Li, A. K. Hopkins, D. Bau, F. Viégas, H. Pfister, and M. Wattenberg, “Emergent world representations: Exploring a sequence model trained on a synthetic task,” in *ICLR*, 2023.
- [2] N. Nanda, A. Lee, and M. Wattenberg, “Emergent linear representations in world models of self-supervised sequence models,” in *BlackboxNLP*, 2023.
- [3] T. McGrath et al., “Acquisition of chess knowledge in AlphaZero,” *PNAS*, vol. 119, no. 47, 2022.
- [4] A. Karvonen, “Emergent world models and latent variable estimation in chess-playing language models,” in *CoLM*, 2024.
- [5] A. F. Spies, W. Edwards, M. I. Ivanitskiy, et al., “Transformers use causal world models in maze-solving tasks,” *arXiv:2412.11867*, 2024.
- [6] X. Zhang, “What do world models learn in RL? Probing latent representations in learned environment simulators,” *arXiv:2603.21546*, 2026.
- [7] M. Wei, M. Freeman, C. Donahue, and C. Sun, “Do music generation models encode music theory?,” in *ISMIR*, 2024.
- [8] I. Prokopiou, P. Vikatos, M. Kaliakatsos-Papakostas, T. Giannakopoulos, and T. Stafylakis, “Latent space disentanglement via activation steering for interpretable attribute control in symbolic music generation,” in *EUSIPCO*, 2026.
- [9] J. Thickstun, D. Hall, C. Donahue, and P. Liang, “Anticipatory music transformer,” *TMLR*, 2024.
- [10] C. L. Krumhansl and E. J. Kessler, “Tracing the dynamic changes in the perceived tonal organization in a spatial representation of musical keys,” *Psychological Review*, vol. 89, no. 4, pp. 334–368, 1982.
- [11] J. Hewitt and P. Liang, “Designing and interpreting probes with control tasks,” in *EMNLP*, 2019.